

BASIC SYMBOLS FOR REGISTER TRANSFERS

Symbols	Description	Examples
Capital letters	Denotes a register	MAR, R2
Parentheses ()	Denotes a part of a register	R2(0-7), R2(L)
Arrow $\leftarrow$	Denotes transfer of information	R2 $\leftarrow$ R1
Colon :	Denotes termination of control function	P:
Comma	Separates two micro-operations	A $\leftarrow$ B, B $\leftarrow$ A



CONNECTING REGISTERS

- In a digital system with many registers, it is impractical to have data and control lines to directly allow each register to be loaded with the contents of every possible other registers
- To completely connect n registers $\rightarrow n(n-1)$ lines
- $O(n^2)$ cost
 - This is not a realistic approach to use in a large digital system
- Instead, take a different approach
- Have one centralized set of circuits for data transfer – the **bus**
- Have control circuits to select which register is the source, and which is the destination



BUS AND MEMORY TRANSFER

- **Bus:** It is a path(of a group of wires) over which information is transferred, from any of several sources to any of several destinations.
- From a register to bus: $BUS \leftarrow R$

